

Chapter 3

3.2

- (a) As x increase 1 unit, the percentage of times the starting pitcher pitched a complete game will decrease 6.62%.
- (b) In the linear prediction, when $x = 12$, the estimated percentage becomes -0.1014 (-10.14%), which is below zero. In the logistic regression, when $x = 12$, the estimated percentage is 0.034 (3.4%). The prediction of logistic regression is more plausible since it is between 0 and 1, while the prediction of linear model is below 0, which isn't reasonable.

3.5

- (a) The estimated linear regression model is: $\hat{P}(Y = 1) = -0.145 + 0.323 \times \text{weight}$.

When weight is equal to 0, the estimated probability of $Y = 1$ is -0.145 . As weight increase 1 unit, the estimated probability will increase by 0.323.

At a weight of 5.20 kg, the estimated probability is 1.533. Since 1.533 is larger than 1, it's not reasonable.

- (b) The estimated logistic regression is: $\log\left(\frac{\hat{P}(Y=1)}{1-\hat{P}(Y=1)}\right) = -3.695 + 1.815 \times \text{weight}$.

At a weight of 5.20 kg, $\log\left(\frac{\hat{P}(Y=1)}{1-\hat{P}(Y=1)}\right) = 5.744 \Rightarrow \hat{P}(Y = 1) = \frac{e^{5.744}}{1+e^{5.744}} = 0.9968$.

3.6

- (a) The estimated logistic regression is: $\log\left(\frac{\hat{P}(Y=1)}{1-\hat{P}(Y=1)}\right) = 3.1870 - 0.5901x$. Since the coefficient of x is negative, it means that as x increases, the probability of being a Democrat will decrease.

- (b) The 95% Wald CI for the effect of political ideology is $\hat{\beta} \pm 1.96(SE) = -0.5901 \pm 1.96(0.1564) = (-0.897, -0.284)$. Compared to the profile CI $(-0.916, -0.298)$, Wald CI is shorter.

- (c) The Wald test statistic is $z = \frac{\hat{\beta}}{SE} = -\frac{0.5901}{0.1564} = -3.772$, the p-value is equal to 0.000162, which implies that the effect of political ideology is significant.

- (d) The likelihood-ratio test statistic is $24.7983 - 7.7894 = 17.0089$ which has an approximate χ_1^2 , the p-value is equal to 0.0000372, which implies that the effect of political ideology is significant.

- (e) The number of Fisher scoring iterations is 4, which means that the ML estimates of the parameters done by Fisher scoring required only 4 iterations to converge.

3.8

The estimated logistic regression model is: $\text{logit} [\hat{P}(Y = 1)] = \log \left(\frac{\hat{P}(Y=1)}{1-\hat{P}(Y=1)} \right) = -3.866 + 0.397x$.

The Wald test for effect of snoring: $z = \frac{0.397}{0.05} = 7.945 > z_{0.975} = 1.96$, which implies the snoring effect is significant.

The Wald 95% CI of the effect of snoring is $\hat{\beta} \pm 1.96(SE) = 0.397 \pm 1.96(0.05) = (0.299, 0.495)$.

3.11

The estimated Poisson regression model is: $\log(\hat{\mu}) = 1.6094 + 0.5878x$.

$e^{0.5878} = 1.8$, which implies that treatment B has an 80.0% increase in the estimated mean number of imperfections.

3.12

(a) Hypothesis test: $\begin{cases} H_0: \beta = 0 \\ H_1: \beta \neq 0 \end{cases}$.

Wald test: $z = \frac{0.5878}{0.1764} = 3.332 > z_{0.975} = 1.96$.

Likelihood-ratio test: $2(L_1 - L_0) = 27.857 - 16.268 = 11.589 > \chi^2_{(1),0.95} = 3.84$.

Since the Wald test and likelihood-ratio test are both significant, it implies treatment A is significantly better than treatment B.

(b)

The Wald 95% CI for β is $\hat{\beta} \pm 1.96(SE) = 0.5878 \pm 1.96(0.1764) = (0.242, 0.934)$. The Wald 95% CI for μ_B/μ_A is $(e^{0.242}, e^{0.934}) = (1.274, 2.544)$.

The profile likelihood 95% CI for β is $(0.2469, 0.9401)$. The profile likelihood 95% CI is $(e^{0.4597}, e^{0.7145}) = (1.28, 2.56)$.

3.13

(a) $\log(\mu) = -0.428 + 0.5893x$. When $x = 2.44$, the estimated mean response is 2.744.

(b) $\hat{\beta} = 0.5893$, $e^{0.5893} = 1.803$, which means a 1-cm increase in the weight has an 80.3% increase in the estimated mean number of satellites.

The Wald 95% CI for β is $\hat{\beta} \pm 1.96(SE) = 0.5893 \pm 1.96(0.065) = (0.4619, 0.7167)$. the Wald 95% CI for the multiplicative effect of a 1-kg increase is $(e^{0.4619}, e^{0.7167}) = (1.587, 2.048)$.

The profile likelihood 95% CI for β is $(0.4597, 0.7145)$. The profile likelihood 95% CI for the multiplicative

effect of a 1-kg increase is $(e^{0.4597}, e^{0.7145}) = (1.584, 2.043)$.

(c)

The Wald test statistic is $z = \frac{\hat{\beta}}{SE} = \frac{0.5893}{0.065} = 9.064 > z_{0.975} = 1.96$.

The likelihood-ratio test is $2(L_1 - L_0) = 632.79 - 560.87 = 71.92 > \chi^2_{(1),0.95} = 3.84$.

Since the Wald test and likelihood-ratio test are both significant, it implies the effect of weight is significant.

3.15

(a) True. (b) False, the variance of Y does not need to be constant at all values of explanatory variables.